

ONTOLOGIE SEMANTICĂ PENTRU BERE

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SCOPUL PROIECTULUI

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Dezvoltarea unei ontologii dedicate berii pentru a facilita căutarea, recomandarea și analiza tipurilor de bere, pornind de la caracteristici precum stilul, ingredientele, procesul de fabricare, conținutul de alcool și rating-urile utilizatorilor.

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ARCHITECTURA

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Date CSV

Conversie OWL

GraphDB

Interogari SPARQL

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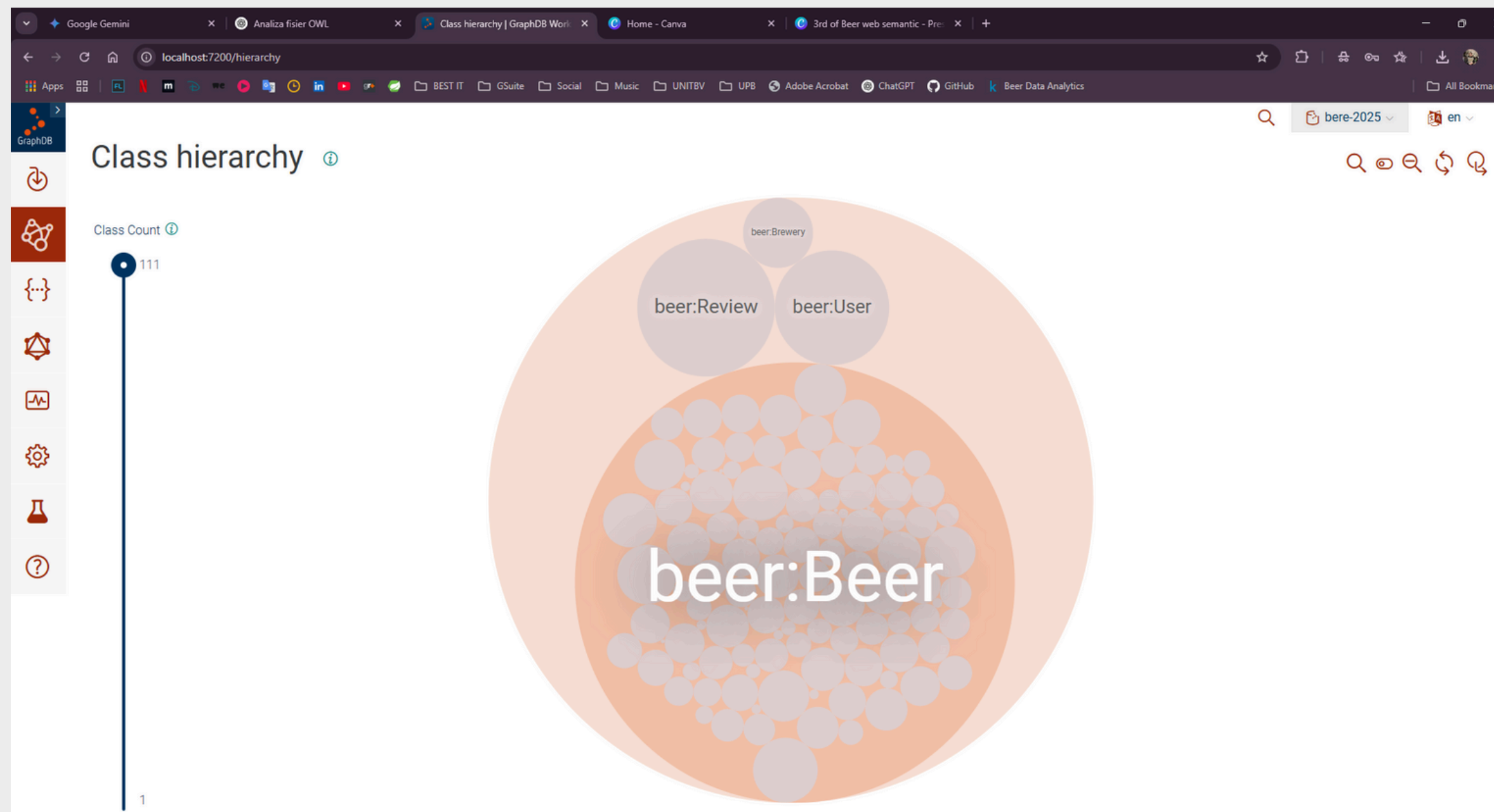
DATE CSV

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500k randuri cu date despre bere

beer_ABV	beer_beerId	beer_brewerId	beer_name	beer_style	review_appearance	review_palette	review_overall	review_taste	review_profileName	review_aroma	review_text (scrut
5.0	47986	10325	Sausa Weizen	Hefeweizen	2.5	2	1.5	1.5	stcules	1.5	A lot of foam.... lactic
6.2	48213	10325	Red Moon	English Strong Ale	3.0	2.5	3.0	3.0	stcules	3.0	Dark red color... caramel bitterness.
6.5	48215	10325	Black Horse Black Beer	Foreign / Export Stout	3.0	2.5	3.0	3.0	stcules	3.0	Almost totally black
5.0	47969	10325	Sausa Pils	German Pilsener	3.5	3	3.0	2.5	stcules	3.0	Golden yellow... grass
7.7	64883	1075	Cauldron DIPA	American Double / Imperial IPA	4.0	4.5	4.0	4.0	johnmichaels	4.5	Orange copper color... hoppy/bitter... very i

CE E DIFERIT?



INTEROGARI SPARQL

```
FILTER (
    ?specificStyle = beer:American_IPA ||
    ?specificStyle = beer:American_Double__Imperial_IPA ||
    ?specificStyle = beer:Belgian_IPA ||
    ?specificStyle = beer:English_India_Pale_Ale_IPA ||
    ?specificStyle = beer:IPA
) .

?beer beer:hasABV ?abv .
?beer beer:hasDescriptiveName ?descriptiveBeerName .
FILTER (?abv > 4.0)
}
LIMIT 10 # Increased limit just in case there are more
```

descriptiveBeerName	abv	specificStyle
"Nick's Balls Of Gold"	"7.2"^^xsd:float	ex:American_IPA
"Caldera IPA"	"6.1"^^xsd:float	ex:American_IPA
"Point Defiance IPA"	"6.1"^^xsd:float	ex:American_IPA
"Fear American IPA"	"7.0"^^xsd:float	ex:American_IPA
"Brass Lantern I.P.A."	"5.8"^^xsd:float	ex:American_IPA
"Hoppin' To Heaven IPA"	"6.8"^^xsd:float	ex:American_IPA

```
SELECT ?descriptiveBeerName ?appearanceScore
WHERE {
    ?beer beer:hasAppearanceScore ?appearanceScore .
    ?beer beer:hasDescriptiveName ?descriptiveBeerName .
    FILTER (?appearanceScore >= 4.0)
}
ORDER BY DESC(?appearanceScore)
LIMIT 10
```

descriptiveBeerName	
"Blackwater Uber Pale Ale"	"5.0"^^xsd:float
"Barrel Aged Naked Evil"	"5.0"^^xsd:float
"Founders Nemesis 2010"	"5.0"^^xsd:float
"Founders Porter"	"5.0"^^xsd:float
"Founders Looking Glass"	"5.0"^^xsd:float
"Founders Devil Dancer"	"5.0"^^xsd:float
"Founders ..."	"5.0"^^xsd:float



THANK YOU